

The Baacke renormalization procedure

Computational steps only: from the divergent object to $\ln D_{\text{ren}}$
for N coupled scalar fields

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Contents

1	The objects	2
2	The starting point: the divergent object	3
3	Step 1 — the full h-ODE	4
4	Step 2 — the linearized Born chain	4
5	Step 3 — the traces $\bar{G}_{(n)}^{(1)}$ and $\bar{G}_{(n)}^{(2)}$	6
5.1	The coincident Green's function	6
5.2	The two trace formulas	6
6	Step 4 — the pointwise subtraction	7
7	Step 5 — dimensional regularization: isolating the two poles	8
7.1	The two Born amplitudes, covariantly	8
7.2	The tadpole integral	8
7.3	The fish integral	8
8	Step 6 — $A_{\text{finite}}^{(1)}(\mu)$, the tadpole addback	10
9	Step 7 — $A_{\text{finite}}^{(2)}(\mu)$, the fish addback	11
9.1	Ingredient 1 — the Fourier transform	11
9.2	Ingredient 2 — reducing $\int d^4p$	11
9.3	Ingredient 3 — assembling	11
9.4	The Feynman-parameter integral in closed form	12
9.5	Truncating the p -integral	12
10	Step 8 — the assembly	13
10.1	Term by term	13
10.2	The dangerous modes, computationally	13
10.3	What is <i>not</i> inside $\ln D_{\text{ren}}$	14
11	Numerical recipe	15
11.1	A worked ledger	15
11.2	Equivalently: bulk + addback	16
12	The check that validates the two addbacks	17
13	Checklist	18

1 The objects

Everything that enters $\ln D_{\text{ren}}$, defined once. i, j, k are *field* indices ($1..N$), α, β, γ are *solution* indices, n is the $O(4)$ principal angular momentum with multiplicity $(n+1)^2$, and $\nu \equiv n+1$.

Object	Definition
<i>Input</i>	
$\bar{\phi}_b(r)$	the bounce profile, N components, in the FV mass eigenbasis
$\bar{\phi}_{(\bar{v})}$	the false vacuum; the origin of that basis
$[U''(\bar{\phi})]_{ij}$	the Hessian of the potential, $\partial^2 U / \partial \phi_i \partial \phi_j$
m_i^2	the FV mass eigenvalues, $= \text{diag} [U''(\bar{\phi}_{(\bar{v})})]_{ij}$
$[U''_{\text{int}}(\bar{\phi}_b(r))]_{ij}$	the interaction Hessian , $[U''(\bar{\phi}_b(r))]_{ij} - m_i^2 \delta_{ij}$. Vanishes identically at the FV. <i>This one object is what every integral below is built from.</i>
μ	the $\overline{\text{MS}}$ renormalization scale (pipeline: $\mu = \sum_i m_i$)
<i>Operators and mode functions</i>	
$[M(x, s^2)]_{ij}$	$[-\Delta^{(4)} + m_i^2 + s^2] \delta_{ij} + [U''_{\text{int}}(\bar{\phi}_b(x))]_{ij}$
$[M_{(\bar{v})}(x, s^2)]_{ij}$	$[-\Delta^{(4)} + m_i^2 + s^2] \delta_{ij}$
s^2	the resolvent parameter; ε^2 , Λ^2 its lower/upper limits
k_i	$\sqrt{m_i^2 + s^2}$
$B_{n+1}^\pm(k_i r)$	the free radial basis: $B^- = I_{n+1}$ (regular at 0), $B^+ = K_{n+1}$ (decaying at ∞)
$\phi_{n,i}^{\alpha,\pm}$	the radial mode functions, $= B_{n+1}^\pm(k_i r) [\delta_i^\alpha + h_{n,i}^{\alpha,\pm}]$
$h_{n,i}^{\alpha,\pm}(r, s^2)$	the deviation functions — the only thing solved for
$h^{(1)}, h^{(2)}, h^{(R)}$	the $O(U''_{\text{int}})$, $O((U''_{\text{int}})^2)$ and $O((U''_{\text{int}})^{\geq 3})$ parts of h
$\mathcal{L}_i^\pm$	the linear operator on the left of the h -ODE, eq. (6)
$C_{\alpha\beta}$	the constant Wronskian, $= r^3 W_{\alpha\beta}(r)$; numerically $\mathbb{I} + h_n^-(\infty)$
<i>Traces — the integrands</i>	
$\bar{G}(s^2)$	$\text{Tr}[M^{-1}] = \sum_i \sum_n \int dr r^3 (n+1)^2 [G_n(r, r, s^2)]_{ii}$
$\bar{G}_{(\bar{v})}(s^2)$	the same on the FV side (closed form: pure Bessel)
$\bar{G}''(s^2)$	$\bar{G}$ with the negative mode and the four zero modes removed
$\bar{G}_{(n)}^{(1)}(s^2)$	the tadpole trace : the $O(U''_{\text{int}})$ part of $\bar{G}_{(n)} - \bar{G}_{(n),(\bar{v})}$, eq. (14)
$\bar{G}_{(n)}^{(2)}(s^2)$	the fish trace : the $O((U''_{\text{int}})^2)$ part, eq. (15)
$\delta_{(n)}^{(\geq 3)}(s^2)$	the graded residual — what is actually integrated, eq. (16)
<i>Counterterm objects</i>	
$[\tilde{U}''_{\text{int}}]_{ij}(p)$	the Fourier transform of $[U''_{\text{int}}]_{ij}(x)$, eq. (33)
Δ_{ij}	$x(1-x)p^2 + x m_j^2 + (1-x) m_i^2$
$\Pi_{ij}(p^2)$	the two-mass bubble, eq. (25)
Π_{ij}^{finite}	its $\overline{\text{MS}}$ finite part, eq. (26)
$A_{\text{finite}}^{(1)}(\mu)$	the tadpole addback , eq. (30)
$A_{\text{finite}}^{(2)}(\mu)$	the fish addback , eq. (34)
$1/\bar{\varepsilon}$	$1/\epsilon - \gamma_E + \ln 4\pi$
Ω_3	$2\pi^2$, the unit 3-sphere area; $\int d^4x f(r) = \Omega_3 \int dr r^3 f(r)$
<i>Mode bookkeeping (enters the assembly, not the renormalization)</i>	
$\omega_{0,0}^2$	the negative eigenvalue, $(L, n) = (0, 0)$, multiplicity 1
$\omega_{0,1}^2$	the zero-mode eigenvalue, $(L, n) = (0, 1)$, multiplicity $(n+1)^2 = 4$
<i>Output</i>	

Object	Definition
$\mathfrak{J}_{\text{ren}}'' = \ln D_{\text{ren}}$	the renormalized log-determinant ratio, eq. (37)

2 The starting point: the divergent object

What is being computed

$$\ln D_{\text{ren}} = \mathfrak{J}_{\text{ren}}'', \quad \mathfrak{J}''(s^2=0) = \ln \frac{\det'' [M(x, 0)]_{ij}}{\det [M_{(\bar{v})}(x, 0)]_{ij}}, \quad (1)$$

obtained *not* as a determinant but as an s^2 -integral of a trace, using $\frac{d}{ds^2} \ln \det M = \text{Tr}[M^{-1}] = \bar{G}(s^2)$:

$$\mathfrak{J}''(0) = - \int_0^\infty ds^2 \left[\bar{G}''(s^2) - \bar{G}_{(\bar{v})}(s^2) \right]. \quad (2)$$

The problem, stated in one line

Per partial wave the integral in (2) is finite. **The sum over n is not.** Organizing $\bar{G} - \bar{G}_{(\bar{v})}$ by powers of U_{int}'' (the Born series), the s^2 -integrated n -th partial wave of Born order k behaves as

$$I_n^{(k)} \propto (n+1)^{-(2k-3)}, \quad (3)$$

so

$$k=1 \text{ (tadpole)} : \sum_n \nu \sim n_{\text{max}}^2 \text{ **quadratic**;} \quad k=2 \text{ (fish)} : \sum_n \nu^{-1} \sim \ln n_{\text{max}} \text{ **log**};$$

$$k \geq 3 : \sum_n \nu^{-3} \text{ **finite**.}$$

Only $k=1$ and $k=2$ diverge. Everything else converges like a Hurwitz zeta.

The recipe, in five moves

1. Solve the **full h -ODE** for $h_{n,i}^{\alpha,\pm}$ (§3).
2. Solve the **linearized chain** for $h^{(1)}$ and $h^{(2)}$ (§4).
3. Build the traces $\bar{G}_{(n)}^{(1)}$, $\bar{G}_{(n)}^{(2)}$ and **subtract them pointwise in (n, s^2)** . The n -sum now converges (§5–§6).
4. That subtraction removed the divergence *and* the finite $\overline{\text{MS}}$ remainder. Compute the two finite parts analytically in dim reg (§7–§9) and **add them back**.
5. Assemble (§10).

3 Step 1 — the full h -ODE

The radial mode equation $[M_n]_{ij}\phi_{n,j}^{\alpha,\pm} = 0$, with $\phi_i = u_i/r$ and $1 + n(n+2) = (n+1)^2$, is the modified Bessel equation of order $\nu = n+1$ with a source:

$$u_i'' + \frac{1}{r}u_i' - \left[\frac{(n+1)^2}{r^2} + k_i^2 \right] u_i = [U_{\text{int}}'']_{ij} u_j, \quad k_i^2 = m_i^2 + s^2. \quad (4)$$

Factor out the free behaviour:

$$\phi_{n,i}^{\alpha,\pm}(r, s^2) = B_{n+1}^\pm(k_i r) \left[\delta_i^\alpha + h_{n,i}^{\alpha,\pm}(r, s^2) \right]. \quad (5)$$

The full h -ODE — everything below is built from its solutions

$$\underbrace{\left\{ \frac{d^2}{dr^2} + \left[\frac{1}{r} + 2k_i \frac{(B_{n+1}^\pm(k_i r))'}{B_{n+1}^\pm(k_i r)} \right] \frac{d}{dr} \right\}}_{\equiv \mathcal{L}_i^\pm} h_{n,i}^{\alpha,\pm}(r, s^2) = [U_{\text{int}}''(\bar{\phi}_b(r))]_{ij} \frac{B_{n+1}^\pm(k_j r)}{B_{n+1}^\pm(k_i r)} \left[\delta_j^\alpha + h_{n,j}^{\alpha,\pm}(r, s^2) \right], \quad (6)$$

with

$$\text{I) } h_{n,i}^{\alpha,-}(0) = (h_{n,i}^{\alpha,-})'(0) = 0, \quad \text{II) } h_{n,i}^{\alpha,+}(\infty) = (h_{n,i}^{\alpha,+})'(\infty) = 0. \quad (7)$$

Conventions. (i) The prime in $(B_{n+1}^\pm(k_i r))'$ is d/dz with respect to the *argument* $z = k_i r$ — hence the explicit k_i . (ii) $B_{n+1}^\pm$ is the *bare* modified Bessel function; absorbing the $1/r$ Jacobian into the envelope changes nothing, since $2\tilde{B}'/\tilde{B} + \frac{3}{r} = \frac{1}{r} + 2B'/B$.

Why h and not ϕ . The drift coefficient is a *logarithmic derivative* of a Bessel function: smooth and well conditioned even where I_{n+1} and K_{n+1} themselves overflow or underflow. In practice one forms the source ratio as $\exp(\ln B_j - \ln B_i)$ from scaled Bessel functions, so only differences of logs ever appear.

4 Step 2 — the linearized Born chain

Write $h = h^{(1)} + h^{(2)} + h^{(R)}$ and collect powers of U_{int}'' in (6). On the right, the bracket $[\delta_j^\alpha + h_{n,j}^{\alpha,\pm}]$ contributes δ at order 0 and h from order 1, while the prefactor $[U_{\text{int}}'']_{ij}$ carries one power — so order k on the left is driven by order $k-1$ inside the bracket.

The chain

First order (pure source, no h on the right; the bracket gives δ_j^α , so $j = \alpha$):

$$\mathcal{L}_i^\pm h_{n,i}^{(1),\alpha,\pm}(r, s^2) = [U_{\text{int}}''(\bar{\phi}_b(r))]_{i\alpha} \frac{B_{n+1}^\pm(k_\alpha r)}{B_{n+1}^\pm(k_i r)} \quad (8)$$

Second order (same operator, driven by the previous rung):

$$\mathcal{L}_i^\pm h_{n,i}^{(2),\alpha,\pm}(r, s^2) = \sum_j [U_{\text{int}}''(\bar{\phi}_b(r))]_{ij} \frac{B_{n+1}^\pm(k_j r)}{B_{n+1}^\pm(k_i r)} h_{n,j}^{(1),\alpha,\pm}(r, s^2) \quad (9)$$

Remainder (all orders ≥ 3 , resummed exactly — not truncated):

$$\mathcal{L}_i^\pm h_{n,i}^{(R),\alpha,\pm} = \sum_j [U_{\text{int}}'']_{ij} \frac{B_{n+1}^\pm(k_j r)}{B_{n+1}^\pm(k_i r)} (h_{n,j}^{(2),\alpha,\pm} + h_{n,j}^{(R),\alpha,\pm}). \quad (10)$$

Each rung carries the *same* boundary conditions I) / II) as the full equation.

These are three separate ODEs. Nothing is ever solved “summed”.

(8), (9) and (10) are solved **one at a time**, in this order. They share the same operator $\mathcal{L}_i^\pm$ and the same source matrix S , and they are coupled only *one way*:

equation	needs as input	unknown
(8)	nothing (pure source S)	$h^{(1)}$
(9)	$h^{(1)}$, already known	$h^{(2)}$
(10)	$h^{(2)}$, already known	$h^{(R)}$

Because $\mathcal{L}_i^\pm$ and S are shared, it is cheapest to stack them into one state vector of $8N^2$ reals and hand that to the solver in a single call — but that is bookkeeping, not a different equation. Each block still satisfies its own equation and its own copy of the boundary conditions I) / II).

Why the identity $h = h^{(1)} + h^{(2)} + h^{(R)}$ is then stated — and what it is for

The display below is **not an equation being solved**. It is a one-line uniqueness argument. Add the three right-hand sides:

$$\mathcal{L}_i^\pm (h^{(1)} + h^{(2)} + h^{(R)}) = S + S h^{(1)} + S h^{(2)} + S h^{(R)} = S(\mathbb{I} + h^{(1)} + h^{(2)} + h^{(R)}).$$

The right-hand side is (6) with h replaced by the sum. So the sum obeys the *same* ODE and the *same* boundary conditions as h — hence, by uniqueness,

$$h = h^{(1)} + h^{(2)} + h^{(R)} \quad \textbf{exactly, not to some order.} \quad (11)$$

Why this is load-bearing. The “full” object $\bar{G}_{(n)} - \bar{G}_{(n),(\bar{v})}$ of §5 is *built from the sum* $h^{(1)} + h^{(2)} + h^{(R)}$ — the pipeline never feeds a separately-solved full h into the traces. Without (11) you would not know that

$$(\text{object built from the sum}) - \bar{G}_{(n)}^{(1)} - \bar{G}_{(n)}^{(2)}$$

is really the Born- ≥ 3 remainder rather than the remainder of a truncated series. It is also why the third rung $h^{(R)}$ exists at all: it is not decorative, it is what makes the sum add up to the full h .

And it is checked at runtime. The full h is integrated as well (a fourth block), purely so that $|h - (h^{(1)} + h^{(2)} + h^{(R)})|$ can be compared against zero at every wave. Measured worst case 9.4×10^{-15} relative over $n = 15 \dots 80$, $s^2 = 0 \dots 100$; the run aborts above 10^{-8} . That block is used for nothing else.

Why this route and not Feynman diagrams. Computing the tadpole and fish diagrams in momentum space and transforming back would give the same numbers but in a *different regularization* than the partial-wave sum uses. Generating $\bar{G}^{(1)}$ and $\bar{G}^{(2)}$ from the same ODE solver, on the same r -grid, at the same (n, s^2) as the full $\bar{G}$ makes the subtraction exact pointwise, with no regulator mismatch and with discretization errors largely cancelling.

5 Step 3 — the traces $\bar{G}_{(n)}^{(1)}$ and $\bar{G}_{(n)}^{(2)}$

5.1 The coincident Green's function

With the constant Wronskian $C_{\alpha\beta} = r^3 W_{\alpha\beta}(r)$, and writing A for the matrix that normalizes it (fixed unambiguously in (13) below),

$$[G_n(r, r, s^2)]_{ii} = \frac{I_{n+1}(k_i r) K_{n+1}(k_i r)}{r^2} \sum_{\alpha, \beta} \left(\delta_{i\alpha} + [h_n^+]_{i\alpha}(r) \right) A_{\alpha\beta} \left(\delta_{i\beta} + [h_n^-]_{i\beta}(r) \right), \quad (12)$$

i.e. in matrix form $(\mathbb{I} + h_n^+(r)) A (\mathbb{I} + h_n^-(r))^T$, traced on the diagonal. whereas on the FV side, where $h \equiv 0$, $[G_{n,(\bar{v})}(r, r, s^2)]_{ii} = I_{n+1} K_{n+1}/r^2$. **The whole bounce–FV difference is carried by the h -functions.**

The normalizer — get this right or nothing decays

The matrix A contracted between the two brackets is

$$A = \left(\mathbb{I} + h_n^+(0) \right)^{-1} = \left(\mathbb{I} + (h_n^-(\infty))^T \right)^{-1}, \quad (13)$$

with **no transpose in the $\alpha\beta$ slot**. The identity $h_n^{(k),+}(0, s^2) = (h_n^{(k),-}(\infty, s^2))^T$ is Sturm–Liouville conservation of $r^3 W$; in practice one reads the constant off at $r = r_{\max}$, where the $-$ branch has a converged plateau, rather than at the origin where the K -branch is stiff. *Using the transposed matrix instead makes $\delta_{(n)}^{(\geq 3)}$ fail to decay in n or s^2 at all.* Expand $A = \mathbb{I} - M + M^2 - \dots$ with $M \equiv h_n^+(0)$ to get the Born grades.

5.2 The two trace formulas

Multiplying (12) by the multiplicity $(n+1)^2$, summing over the field index and integrating with the measure $r^3 \cdot r^{-2} = r$:

Tadpole trace

$$\bar{G}_{(n)}^{(1)}(s^2) = (n+1)^2 \sum_{i=1}^N \int_0^\infty dr \, r \, I_{n+1}(k_i r) K_{n+1}(k_i r) \left[[h_n^{(1),+}]_{ii}(r) + [h_n^{(1),-}]_{ii}(r) - [h_n^{(1),+}]_{ii}(0) \right] \quad (14)$$

Fish trace (all products are ordinary matrix products, traced on the diagonal)

$$\begin{aligned} \bar{G}_{(n)}^{(2)}(s^2) = (n+1)^2 \sum_{i=1}^N \int_0^\infty dr \, r \, I_{n+1} K_{n+1} \left[h_n^{(2),+} + h_n^{(2),-} - h_n^{(2),+}(0) \right. \\ \left. + h_n^{(1),+} h_n^{(1),-} - h_n^{(1),+} h_n^{(1),+}(0) \right. \\ \left. - h_n^{(1),+}(0) h_n^{(1),-} + (h_n^{(1),+}(0))^2 \right]_{ii} \end{aligned} \quad (15)$$

i.e. in index form the four quadratic terms are

$$\begin{aligned} & + \sum_{\alpha} [h_n^{(1),+}]_{i\alpha}(r) [h_n^{(1),-}]_{i\alpha}(r), \quad - \sum_{\alpha} [h_n^{(1),+}]_{i\alpha}(r) [h_n^{(1),+}]_{\alpha i}(0), \\ & - \sum_{\beta} [h_n^{(1),+}]_{i\beta}(0) [h_n^{(1),-}]_{i\beta}(r), \quad + \sum_{\gamma} [h_n^{(1),+}]_{i\gamma}(0) [h_n^{(1),+}]_{\gamma i}(0). \end{aligned}$$

Where the seven terms come from. Expanding (12) to $O((U''_{\text{int}})^2)$ with $A = \mathbb{I} - M + M^2$: three from $\delta \cdot \delta$ with second-order h and the second-order part of $-M$; one from $h^{(1),+}(r) \delta_{\alpha\beta} h^{(1),-}(r)$; two from $-M^{(1)}$ contracted against one first-order h on either side; one from $M^{(1)}M^{(1)}$.

Gauge. If the solver enforces the unit-Wronskian normalization $h(0, s^2) = 0$, every $h(0)$ term drops and (14)–(15) collapse to $\bar{G}^{(1)} \propto \int dr r IK [h^{(1),+} + h^{(1),-}]_{ii}$ and $\bar{G}^{(2)} \propto \int dr r IK [h^{(2),+} + h^{(2),-} + h^{(1),+}h^{(1),-}]_{ii}$. The production pipeline does *not* use that gauge — it keeps C explicit and inverts it exactly — so the full forms above are the ones to use.

6 Step 4 — the pointwise subtraction

$$\delta_{(n)}^{(\geq 3)}(s^2) \equiv \underbrace{\bar{G}_{(n)}(s^2)}_{\text{full } h} - \underbrace{\bar{G}_{(n),(\bar{v})}(s^2)}_{\text{free}} - \underbrace{\bar{G}_{(n)}^{(1)}(s^2)}_{h^{(1)}} - \underbrace{\bar{G}_{(n)}^{(2)}(s^2)}_{h^{(1),h^{(2)}}}. \quad (16)$$

The order of operations is not negotiable

Subtract **pointwise in** (n, s^2) ; then do the s^2 -integral **per partial wave**; then sum over n :

$$- \sum_n \int_{\varepsilon^2}^{\Lambda^2} ds^2 \delta_{(n)}^{(\geq 3)}(s^2). \quad (17)$$

Not the other way round: summing over n first leaves a divergent integrand at every s^2 and there is nothing to integrate. Done in this order, each wave gives a finite number decaying as ν^{-3} , and the n -sum converges like a Hurwitz zeta.

Two decay laws are used to close the truncations:

$$\text{in } n : I_n(\infty) \equiv - \int_0^\infty ds^2 \delta_{(n)}^{(\geq 3)} \simeq \frac{a}{\nu^3} + \frac{c}{\nu^5} + \frac{e}{\nu^7}, \quad a \leftrightarrow A_3 = \frac{1}{16} \int dr r^5 \text{Tr}(U''_{\text{int}})^3, \quad (18)$$

$$\text{in } s^2 : \delta_{(n)}^{(\geq 3)}(s^2) \sim -\frac{5}{32} \nu^2 \int_0^\infty dr r^7 \text{Tr}(U''_{\text{int}})^3 \chi^{-7}, \quad \chi = \sqrt{\nu^2 + (s^2 + \bar{m}^2)r^2}, \quad (19)$$

i.e. $(s^2)^{-7/2}$ per wave (not the naive Born count $(s^2)^{-3}$), summing to $(s^2)^{-2}$. Only the *shape* (19) is assumed; it is calibrated at the measured endpoint $\delta_{(n)}^{(\geq 3)}(\Lambda^2)$.

7 Step 5 — dimensional regularization: isolating the two poles

The subtraction (16) removed the *whole* $\bar{G}^{(1)} + \bar{G}^{(2)}$. $\overline{\text{MS}}$ removes only the poles. To restore the difference one must know both loops exactly. Two master formulas suffice:

$$\mu^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 + \Delta)^n} = \frac{\mu^{2\epsilon}}{(4\pi)^{d/2}} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \Delta^{\frac{d}{2}-n}, \quad \frac{1}{AB} = \int_0^1 dx \frac{1}{[xA + (1-x)B]^2}, \quad (20)$$

with $d = 4 - 2\epsilon$ and $1/\bar{\epsilon} \equiv 1/\epsilon - \gamma_E + \ln 4\pi$.

7.1 The two Born amplitudes, covariantly

From $\mathfrak{J}''(0) = \text{Tr} \ln[\mathbb{I} + X] = \sum_k \frac{(-1)^{k+1}}{k} \text{Tr}[X^k]$ with $X_{ij} = [U''_{\text{int}}]_{ik} \bar{G}_{(\bar{v})}(x, x', m_k^2) \delta_{kj}$:

$$A^{(1)} = \text{Tr} \left[\bar{G}_{(\bar{v})}(x, x', m_i^2) [U''_{\text{int}}(\bar{\phi}_b(x'))]_{ij} \right] = \sum_i \left[\int \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2 + m_i^2} \right] \Omega_3 \int_0^\infty dr r^3 [U''_{\text{int}}]_{ii}, \quad (21)$$

$$A^{(2)} = -\frac{1}{2} \text{Tr} \left[\left(\bar{G}_{(\bar{v})}(x, x', m_i^2) \delta_{ik} [U''_{\text{int}}]_{kj}(x, x') \right)^2 \right] = -\frac{1}{2} \sum_{i,j} \int \frac{d^4 p}{(2\pi)^4} |[\tilde{U}''_{\text{int}}]_{ij}(p)|^2 \Pi_{ij}(p^2). \quad (22)$$

The tadpole sees only the diagonal $[U''_{\text{int}}]_{ii}$; the fish sees the full matrix, including the cross-couplings $i \neq j$ with two different masses in the loop.

7.2 The tadpole integral

Apply (20) with $n = 1$, $\Delta = m_i^2$, $d/2 = 2 - \epsilon$, and $\Gamma(-1 + \epsilon) = -\frac{1}{\epsilon} + \gamma_E - 1 + \dots$:

$$\begin{aligned} \mu^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 + m_i^2} &= \frac{\mu^{2\epsilon} \Gamma(-1 + \epsilon)}{(4\pi)^{2-\epsilon}} (m_i^2)^{1-\epsilon} = \frac{m_i^2}{16\pi^2} \left[-\frac{1}{\epsilon} + \gamma_E - 1 - \ln 4\pi - \ln \frac{\mu^2}{m_i^2} \right] \\ &= -\frac{m_i^2}{16\pi^2} \left[\frac{1}{\bar{\epsilon}} + 1 - \ln \frac{m_i^2}{\mu^2} \right]. \end{aligned} \quad (23)$$

7.3 The fish integral

Combine denominators with $A = (k + p)^2 + m_j^2$, $B = k^2 + m_i^2$:

$$xA + (1-x)B = \underbrace{(k + xp)^2}_{\ell^2} + \underbrace{x(1-x)p^2 + xm_j^2 + (1-x)m_i^2}_{\Delta_{ij}}, \quad (24)$$

then (20) at $n = 2$ with $\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma_E + \dots$:

$$\Pi_{ij}(p^2) = \int_0^1 dx \frac{\mu^{2\epsilon} \Gamma(\epsilon)}{(4\pi)^{2-\epsilon}} \Delta_{ij}^{-\epsilon} = \frac{1}{16\pi^2} \int_0^1 dx \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta_{ij}}{\mu^2} \right], \quad (25)$$

$$\Pi_{ij}^{\text{finite}}(m_i^2, m_j^2, p^2) = -\frac{1}{16\pi^2} \int_0^1 dx \ln \frac{\Delta_{ij}}{\mu^2}. \quad (26)$$

Both poles multiply *local* functionals of the bounce

$$A^{(1)} \Big|_{\text{pole}} = -\frac{1}{16\pi^2} \frac{1}{\bar{\epsilon}} \sum_i m_i^2 \Omega_3 \int_0^\infty dr r^3 [U''_{\text{int}}]_{ii} \quad \propto \int d^4 x \text{Tr} [m^2 U''_{\text{int}}], \quad (27)$$

$$A^{(2)}\Big|_{\text{pole}} = -\frac{1}{32\pi^2} \frac{1}{\bar{\varepsilon}} \int d^4x \operatorname{Tr} \left[(U''_{\text{int}}(\bar{\phi}_b(x)))^2 \right] \propto \int d^4x \operatorname{Tr} [(U''_{\text{int}})^2], \quad (28)$$

(the second by Parseval, $\sum_{ij} \int \frac{d^4p}{(2\pi)^4} |\tilde{U}''_{\text{int}}]_{ij}|^2 = \int d^4x \operatorname{Tr} [(U''_{\text{int}})^2]$; the pole of Π_{ij} is p -independent so it factors out). Both operators are built from U''_{int} , which vanishes at the false vacuum — hence $S^{CT}(\bar{\phi}_{(\bar{v})}) = 0$ and only the bounce contributes.

Caveat: do **not** read this as “tadpole = δm^2 , fish = $\delta\lambda$ ”. Born order and field power are orthogonal axes: $\int \operatorname{Tr}[U''_{\text{int}}]$ spreads over χ^1, χ^2 and $\int \operatorname{Tr}[(U''_{\text{int}})^2]$ over χ^2, χ^3, χ^4 (with $\chi = \bar{\phi}_b - \bar{\phi}_{(\bar{v})}$), so δm^2 collects from both. Only the total is axis-independent.

8 Step 6 — $A_{\text{finite}}^{(1)}(\mu)$, the tadpole addback

Take the finite part of (23) and multiply by the radial factor of (21), using $\int d^4x = \Omega_3 \int_0^\infty dr r^3$ with $\Omega_3 = 2\pi^2$:

$$A_{\text{finite}}^{(1)}(\mu) = \sum_i \left(-\frac{m_i^2}{16\pi^2} \right) \left[1 - \ln \frac{m_i^2}{\mu^2} \right] \cdot 2\pi^2 \int_0^\infty dr r^3 [U_{\text{int}}''(\bar{\phi}_b(r))]_{ii}, \quad \frac{2\pi^2}{16\pi^2} = \frac{1}{8}. \quad (29)$$

Result — fully explicit, nothing left to solve

$$A_{\text{finite}}^{(1)}(\mu) = -\frac{1}{8} \sum_{i=1}^N m_i^2 \left[1 - \ln \frac{m_i^2}{\mu^2} \right] \int_0^\infty dr r^3 [U_{\text{int}}''(\bar{\phi}_b(r))]_{ii} \quad (30)$$

Ingredients: the FV masses m_i^2 , the scale μ , and **one radial quadrature** of the diagonal of the interaction Hessian. No loop integral, no ODE, no partial-wave sum. The N -field structure enters only as $\sum_i$.

The 1/8 is $\Omega_3/(16\pi^2)$, not a symmetry factor — the 3-sphere area over the universal one-loop $16\pi^2$. The same Ω_3 gives the fish its $1/(8\pi^2) = \Omega_3/(2\pi)^4$.

Basis condition

(30) is valid *only in the FV mass eigenbasis*: it keeps the diagonal of U_{int}'' because $\bar{G}_{(\bar{v})}(x, x, m_i^2) \delta_{ij}$ is diagonal only there. The same applies to Π_{ij} , which assigns m_i^2, m_j^2 per channel index. Rotate the bounce into that frame first.

Reference implementation

```
A1 = 0.0
for i in range(N):
    # sum over ALL N channels
    I_R = trapezoid(R**3 * U_int_diag[i], R) # Int dr r^3 [U''_int]_ii
    log_factor = 1.0 - np.log(m_sq[i] / mu**2) # 1 - ln(m_i^2/mu^2)
    A1 += m_sq[i] * log_factor * I_R
A1_fin = -A1 / 8.0
```

9 Step 7 — $A_{\text{finite}}^{(2)}(\mu)$, the fish addback

Three ingredients: the Fourier transform of U''_{int} , the reduction of $\int d^4p$, and Π^{finite} .

9.1 Ingredient 1 — the Fourier transform

$$\begin{aligned} [\tilde{U}''_{\text{int}}]_{ij}(p) &= \int d^4x e^{-ipx} [U''_{\text{int}}]_{ij}(x) = \int dr r^3 4\pi \int_0^\pi \sin^2\Theta d\Theta e^{-ipr \cos\Theta} [U''_{\text{int}}]_{ij}(r) \\ &= \int dr r^3 4\pi \int_0^\pi \sin^2\Theta \cos(pr \cos\Theta) d\Theta [U''_{\text{int}}]_{ij}(r) \end{aligned} \quad (31)$$

(the sin part is odd in $\cos\Theta$ and drops; $\int d\Omega_3 f(\Theta) = 4\pi \int_0^\pi \sin^2\Theta f d\Theta$). The angular integral is a Bessel function: from the Poisson representation at $\nu = 1$, with $\pi^{1/2}\Gamma(\frac{3}{2}) = \frac{\pi}{2}$,

$$\int_0^\pi \sin^2\Theta \cos(z \cos\Theta) d\Theta = \frac{\pi J_1(z)}{z}. \quad (32)$$

Fourier transform

$$[\tilde{U}''_{\text{int}}]_{ij}(p) = 4\pi^2 \int_0^\infty dr r^3 \frac{J_1(pr)}{pr} [U''_{\text{int}}(\bar{\phi}_b(r))]_{ij} \quad (33)$$

Use the exact limit $J_1(z)/z \rightarrow \frac{1}{2}$ as $z \rightarrow 0$ (DLMF 10.7.3) for the $pr \rightarrow 0$ column.

Built-in check: at $p = 0$ this gives $[\tilde{U}''_{\text{int}}]_{ij}(0) = 4\pi^2 \cdot \frac{1}{2} \int dr r^3 [U''_{\text{int}}]_{ij} = \Omega_3 \int dr r^3 [U''_{\text{int}}]_{ij}$ — exactly the geometric factor of the tadpole (21). *The tadpole is the fish's Fourier transform at zero momentum, with one propagator instead of two.* ✓

9.2 Ingredient 2 — reducing $\int d^4p$

$[\tilde{U}''_{\text{int}}]_{ij}$ depends only on $|p|$, so $\int \frac{d^4p}{(2\pi)^4} = \frac{\Omega_3}{(2\pi)^4} \int_0^\infty dp p^3$ with $\frac{\Omega_3}{(2\pi)^4} = \frac{2\pi^2}{16\pi^4} = \frac{1}{8\pi^2}$.

9.3 Ingredient 3 — assembling

Result

$$A_{\text{finite}}^{(2)}(\mu) = -\frac{1}{2} \sum_{i,j=1}^N \frac{\Omega_3}{(2\pi)^4} \int_0^\infty dp p^3 |[\tilde{U}''_{\text{int}}]_{ij}(p)|^2 \Pi_{ij}^{\text{finite}}(m_i^2, m_j^2, p^2) \quad (34)$$

Pulling out $-\frac{1}{2} \cdot \frac{1}{8\pi^2} = -\frac{1}{16\pi^2}$, the $N = 2$ case reads

$$\begin{aligned} A_{\text{finite}}^{(2)}(\mu) &= -\frac{1}{16\pi^2} \int_0^\infty dp p^3 \left[|[\tilde{U}''_{\text{int}}]_{11}|^2 \Pi^{\text{fin}}(m_1^2, m_1^2, p^2) \right. \\ &\quad + |[\tilde{U}''_{\text{int}}]_{22}|^2 \Pi^{\text{fin}}(m_2^2, m_2^2, p^2) \\ &\quad \left. + 2 |[\tilde{U}''_{\text{int}}]_{12}|^2 \Pi^{\text{fin}}(m_1^2, m_2^2, p^2) \right]. \end{aligned} \quad (35)$$

The factor 2 is the $(1,2) + (2,1)$ pair; for general N simply run the *unrestricted* double sum $\sum_{i,j=1}^N$ — the two orderings are equal because U''_{int} is symmetric and Π^{fin} is symmetric under $i \leftrightarrow j$ via $x \rightarrow 1 - x$.

9.4 The Feynman-parameter integral in closed form

For $m_i = m_j = m$, with $\Delta = x(1-x)p^2 + m^2$,

$$\Pi^{\text{finite}}(m^2, m^2, p^2) = -\frac{1}{16\pi^2} \left[\ln \frac{m^2}{\mu^2} - 2 + \frac{\sqrt{p^2 + 4m^2}}{p} \ln \left(\frac{\sqrt{p^2 + 4m^2} + p}{\sqrt{p^2 + 4m^2} - p} \right) \right], \quad (36)$$

verified against the x -quadrature to 10^{-18} over $p \in [0.3, 40]$, $m \in \{1, 2.6\}$. At $\mu = m$ the bracket becomes $2 - \frac{\sqrt{p^2 + 4m^2}}{p} \ln \frac{\sqrt{p^2 + 4m^2} + p}{\sqrt{p^2 + 4m^2} - p}$, which is exactly $16\pi^2 \Pi^{\text{finite}}$ and is the bracket printed in the single-field literature. Π_{ij}^{finite} with Δ_{ij} is the N -field generalization of that bracket. For $m_i \neq m_j$ evaluate (26) as a one-dimensional quadrature.

Reference implementation

```
# 1. Hankel transform, precomputed on the whole (p,r) grid
PR = np.outer(p_grid, R)
Kmat = np.where(PR > 1e-10, jv(1, PR) / PR, 0.5) # J_1(pr)/(pr), limit 1/2
U_tilde[:,i,j] = (2*np.pi)**2 * trapezoid(R**3 * U_int[:,i,j] * Kmat, R)

# 2. Feynman parameter, x_grid = linspace(0,1,200)
Delta = x*(1-x)*p**2 + x*mj2 + (1-x)*mi2
Pi_fin = -(1/(16*np.pi**2)) * trapezoid(np.log(Delta/mu**2), x_grid)

# 3. p-integral, unrestricted double sum over all N^2 channel pairs
full_int[ip] += p**3 * U_tilde[ip,i,j]**2 * Pi_fin
A2_bare = trapezoid(full_int, p_grid) / (8.0*np.pi**2) # NOTE: no -1/2 here
A2_finite = -0.5 * A2_bare
```

The $-\frac{1}{2}$ normalization trap

The natural code variable is $A2_bare = \frac{1}{8\pi^2} \int dp p^3 \sum_{ij} |\tilde{U}''_{\text{int}}|_{ij}|^2 \Pi_{ij}^{\text{finite}}$, which **omits** the $-\frac{1}{2}$ of (34). Then $A_{\text{finite}}^{(2)} = -\frac{1}{2} A2_bare$, and an assembly line reading $\dots + A1_fin - 0.5*A2_bare$ is adding $A_{\text{finite}}^{(2)}$ with a **plus**. It is not a sign flip. Three conventions in circulation:

	fish variable	assembly
Baacke / literature	$A_{\text{fin}}^{(2)}$: bare graph, no $-\frac{1}{2}$	$S_{\text{eff}} = \frac{1}{2} (A_{\text{fin}}^{(1)} - \frac{1}{2} A_{\text{fin}}^{(2)})$
this note	$A_{\text{finite}}^{(2)} = -\frac{1}{2} A_{\text{fin}}^{(2),\text{BL}}$	$\mathfrak{J}''_{\text{ren}} = \dots + A_{\text{finite}}^{(1)} + A_{\text{finite}}^{(2)}$
pipeline	$A2_fin = A_{\text{fin}}^{(2),\text{BL}}$	$\text{lnD_ren} = \dots + A1_fin - 0.5*A2_fin$

Consequently $\ln D_{\text{ren}} = 2 S_{\text{eff}}^{1-\text{loop}}(\text{Baacke})$, and $A_{\text{finite}}^{(2)}$ is typically *positive* where Baacke's is negative.

9.5 Truncating the p -integral

$\int_0^\infty dp$ must be cut at some P . Because $p^3 |\tilde{U}''_{\text{int}}(p)|^2$ decays fast (smooth localized insertion) while Π^{finite} grows only like $\ln p^2$, a ladder converges quickly: start at $P = 30$, increase in steps of 10, stop when the relative change falls below 10^{-6} , holding the grid spacing dp fixed as P grows so the low- p region keeps its resolution. A *fixed* P is not safe — $[\tilde{U}''_{\text{int}}]_{ij}$ decays on the scale of the wall thickness, which is bounce-specific.

10 Step 8 — the assembly

The final formula

$$\begin{aligned}
\ln D_{\text{ren}} = \mathfrak{J}_{\text{ren}}'' &= - \int_{\varepsilon^2}^{\Lambda^2} ds^2 \left[\underbrace{\bar{G}''(s^2) - \bar{G}_{(\bar{v})}(s^2) - \bar{G}^{(1)}(s^2) - \bar{G}^{(2)}(s^2)}_{\text{dangerous poles removed}} \right] \\
&= \sum_n \delta_{(n)}^{(\geq 3)}(s^2) \text{ (dangerous poles removed)} \\
&\quad + \underbrace{\ln |\omega_{0,0}^2|}_{\text{negative mode}} + \underbrace{A_{\text{finite}}^{(1)}(\mu) + A_{\text{finite}}^{(2)}(\mu)}_{\text{the addbacks}}. \tag{37}
\end{aligned}$$

10.1 Term by term

Term	Sign	What it is
$\bar{G}''$	$-\int$	the full bounce-side trace from the full h -ODE, dangerous modes removed. The only expensive object.
$-\bar{G}_{(\bar{v})}$	$+\int$	the free trace. Closed form (Bessel); in practice subtracted <i>pointwise in r</i> inside the bracket, never formed separately.
$-\bar{G}^{(1)}$	$+\int$	tadpole, from $h^{(1)}$ (8). Removes the quadratic divergence of $\sum_n$.
$-\bar{G}^{(2)}$	$+\int$	fish, from $h^{(1)}, h^{(2)}$ (9). Removes the log divergence.
$+\ln \omega_{0,0}^2 $	$+$	negative-mode addback: its pole was subtracted to make the s^2 -integral converge, but the mode does contribute $ \omega_{0,0}^2 ^{-1/2}$ to the rate.
$+A_{\text{finite}}^{(1)}(\mu)$	$+$	(30): restores the $\overline{\text{MS}}$ finite part of the tadpole that the pointwise subtraction removed along with the divergence.
$+A_{\text{finite}}^{(2)}(\mu)$	$+$	(34): the same for the fish. Carries the cross-couplings.

10.2 The dangerous modes, computationally

Not renormalization, but these objects sit in (37) and must be handled or the s^2 -integral does not exist. $\bar{G}(s^2) = \sum_{L,n} (n+1)^2 / (\omega_{L,n}^2 + s^2)$ has poles at $s^2 = -\omega_\gamma^2$; two lie in the integration range:

mode	(L, n)	mult.	treatment
negative	$(0, 0)$	1	subtract $1/(\omega_{0,0}^2 + s^2)$; add back $\ln \omega_{0,0}^2 $
zero (translations)	$(0, 1)$	4	subtract $4/(\omega_{0,1}^2 + s^2)$; no add-back (amputated)

The asymmetry is physical: the negative mode contributes to the rate, the four zero modes were traded for the collective coordinates $(S_E/2\pi\hbar)^2 VT$ which live *outside* $\ln D_{\text{ren}}$. Net effect after the pole subtraction and the log add-back:

$$n = 0 : \quad \left[- \int_0^{\Lambda^2} ds^2 \delta_{(0)}^{(\geq 3)} \right]_{\text{P.V.}} \quad (\text{negative mode kept, as a principal value}) \tag{38}$$

$$n = 1 : \quad - \int_0^{\Lambda^2} ds^2 \delta_{(1)}^{(\geq 3)} - 4 \ln \lambda_1 \quad (\text{zero modes removed — the det' amputation}) \tag{39}$$

with λ_1 the numerical zero-mode eigenvalue. There is deliberately no $+4 \ln \lambda_1$; it would diverge as $\lambda_1 \rightarrow 0$.

The false-vacuum side is **never projected**, so the denominator is the complete FV determinant and no L_{FV} -type addback appears. (In practice the mode-count balance is restored by inserting five fictitious modes at an arbitrary m_f^2 and removing their determinant again — the two cancel identically for every m_f^2 , leaving $-\ln(\omega_{0,0}^2 + S) - 4\ln(\lambda_1 + S)$.)

10.3 What is *not* inside $\ln D_{\text{ren}}$

The outer $-\frac{1}{2}$, the collective-coordinate factor $(S_E/2\pi\hbar)^2 VT$, S_E itself and $\hbar$. The rate is

$$\frac{\gamma}{V} = \left(\frac{S_E}{2\pi\hbar}\right)^2 e^{-\frac{1}{2} \ln D_{\text{ren}}} e^{-S_E(\bar{\phi}_b)/\hbar}. \quad (40)$$

11 Numerical recipe

Object	Method
$h, h^{(1)}, h^{(2)}, h^{(R)}$	one bundled <code>solve_ivp</code> per branch, LSODA, <code>rtol=1e-11</code> , <code>atol=1e-13</code> , sampled on a fixed r -grid via <code>t_eval</code> (no dense output). The $-$ branch outward from a small r_{start} , the $+$ branch inward from r_{max} , both with $y = 0$ at the start.
$B_{n+1}^{\pm}, I_{n+1}K_{n+1}$	scaled <code>ive/kve</code> ; log-derivatives from the exact recurrences $\frac{d}{dz} \ln I_{\nu} = \frac{\nu}{z} + \frac{I_{\nu+1}}{I_{\nu}}$, $\frac{d}{dz} \ln K_{\nu} = -\frac{\nu}{z} - \frac{K_{\nu-1}}{K_{\nu}}$, with the small- z limits $z/2(\nu \pm 1)$ where the scaled values flush, and the uniform fallback $I_{\nu}K_{\nu} \simeq (2\sqrt{\nu^2 + z^2})^{-1}$.
radial integral	composite trapezoid on a uniform grid; measure $w_i(r) = r I_{n+1}(k_i r) K_{n+1}(k_i r)$.
$\int ds^2$	log-Simpson: Simpson in $u = \ln s^2$ of δs^2 on $[s_{\min}^2, 50]$, linear Simpson on $[50, \Lambda^2]$, plus a rectangle $\delta(s_{\min}^2) s_{\min}^2$ for $[0, s_{\min}^2]$. The overall minus of (37) is applied here.
high- n tail	fit $a/\nu^3 + c/\nu^5 + e/\nu^7$ on a window, sum in closed form: $a\zeta(3, n_{\max}+2) + c\zeta(5, n_{\max}+2) + e\zeta(7, n_{\max}+2)$ (Hurwitz).
s^2 completion	integrate the shape (19) from Λ^2 to ∞ , <i>calibrated</i> at the measured $\delta_{(n)}^{(\geq 3)}(\Lambda^2)$.
$A_{\text{finite}}^{(1)}$	one trapezoid on the bounce grid.
$A_{\text{finite}}^{(2)}$	<code>jv(1, .)</code> Hankel kernel + trapezoid in r ; $x \in [0, 1]$ with ~ 200 points; adaptive p -cutoff ladder.

Three things that bite. (i) The low- s^2 rectangle is legitimate only because the *subtracted* integrand has a finite $s^2 \rightarrow 0$ plateau; applied to the raw sector integrand ($\sim \deg_n / (\lambda_n + s^2)$) it is badly wrong. Integrate after the pole subtraction. (ii) Simpson is not optional: on one production band, log-Simpson -802.396 , linear Simpson -802.350 , trapezoid -805.500 . (iii) Quote only the *summed* tail. c and e are interpolation coefficients ($\sim 10^8$, $\sim -10^{10}$) that move by tens of percent with the fit window while the tail moves by $\sim 2 \times 10^{-3}$; only a has an analytic counterpart, A_3 .

11.1 A worked ledger

Canonical `bdet` bounce, $\mu = 5.198119$, $\Lambda^2 = 825$, $n_{\max} = 255$:

$n = 0$ sector, pole-subtracted	+6.085974
$n = 1$ sector, pole-subtracted (deg. 4)	+24.178594
mode-count balance block	-33.576803
$+\ln \omega_{0,0}^2 $	-2.362311
$\sum_{n=2}^{255}$ band	-804.860617
tail, $n > 255$ (Hurwitz ζ)	-2.576169
$+A_{\text{finite}}^{(1)}(\mu)$	+4823.303891
$+A_{\text{finite}}^{(2)}(\mu)$	-1655.382303
$\ln D_{\text{ren}}$	+2354.810256

Read the magnitudes. The addbacks are $+4823$ and -1655 ; the numerically integrated part is ≈ -813 . The answer is a difference of much larger numbers — the signature of a renormalized

quantity. It also means **the two addbacks must be right to many digits**: a 0.1% error in $A_{\text{finite}}^{(1)}$ alone is ± 5 in $\ln D_{\text{ren}}$. Hence the next section.

11.2 Equivalently: bulk + addback

$$\underbrace{\ln D_{\text{full}}}_{\text{bulk}} + \underbrace{\text{Sub} - \text{CT}(\mu)}_{\text{addback}} = \ln D_{\text{full}} - \text{CT}(\mu) = \ln D_{\text{ren}}. \quad (41)$$

With $\ln D_{\text{full}} = P/\epsilon + F_0$, $\text{CT} = P/\epsilon + c(\mu)$ and $\text{Sub} = P/\epsilon + \sigma$ for *any* finite σ : $\text{bulk} = F_0 - \sigma$, $\text{addback} = \sigma - c(\mu)$, $\text{total} = F_0 - c(\mu)$, independent of σ . In pure $\overline{\text{MS}}$, $c = 0$, so $\ln D_{\text{ren}} = F_0$. **The choice of what to subtract cancels out of the total** — Born loops (here) or WKB large- ℓ asymptotics (Carosi) move the split, not the answer. For the ledger above: $\text{bulk} = -813.111$, $\text{addback} = +3167.922$.

One wording point: the object carrying “divergence *plus* finite remainder” is **the loop you subtract**, not the counterterm. In $\overline{\text{MS}}$ the counterterm is just the pole. That is exactly why an addback is needed.

12 The check that validates the two addbacks

The addbacks are the two largest terms in the ledger, so they need an independent guard. There is a free one, and it is sharp.

(a) From the addbacks. All the μ -dependence of $\ln D_{\text{ren}}$ sits in $A_{\text{finite}}^{(1)} + A_{\text{finite}}^{(2)}$ — the numerically subtracted integral contains no μ at all. Using $\partial_{\ln \mu} \ln(m^2/\mu^2) = -2$ and $\partial_{\ln \mu} \Pi_{ij}^{\text{finite}} = +\frac{1}{8\pi^2}$:

$$\frac{\partial A_{\text{finite}}^{(1)}}{\partial \ln \mu} = \sum_i \left(-\frac{1}{8}\right) m_i^2 \cdot (+2) \int_0^\infty dr r^3 [U_{\text{int}}'']_{ii} = -\frac{1}{4} \int_0^\infty dr r^3 \text{Tr}[m^2 U_{\text{int}}''], \quad (42)$$

$$\frac{\partial A_{\text{finite}}^{(2)}}{\partial \ln \mu} = -\frac{1}{2} \cdot \frac{1}{8\pi^2} \sum_{i,j} \int \frac{d^4 p}{(2\pi)^4} |\tilde{U}_{\text{int}}''(p)|^2 \stackrel{\text{Parseval}}{=} -\frac{1}{16\pi^2} \int d^4 x \text{Tr}[(U_{\text{int}}'')^2] = -\frac{1}{8} \int_0^\infty dr r^3 \text{Tr}[(U_{\text{int}}'')^2]. \quad (43)$$

$$\frac{\partial \ln D_{\text{ren}}}{\partial \ln \mu} = -\frac{1}{4} \int_0^\infty dr r^3 \text{Tr}[m^2 U_{\text{int}}''] - \frac{1}{8} \int_0^\infty dr r^3 \text{Tr}[(U_{\text{int}}'')^2]. \quad (44)$$

(b) From Coleman–Weinberg, independently. With $V_1 = \frac{1}{64\pi^2} \text{Tr}[\mathcal{M}^4 (\ln \frac{\mathcal{M}^2}{\mu^2} - \frac{3}{2})]$, $\mathcal{M}^2 = m^2 + U_{\text{int}}''$, so $\partial_{\ln \mu} V_1 = -\frac{1}{32\pi^2} \text{Tr}[\mathcal{M}^4]$ and $\text{Tr}[\mathcal{M}^4] - \text{Tr}[m^4] = 2\text{Tr}[m^2 U_{\text{int}}''] + \text{Tr}[(U_{\text{int}}'')^2]$. With $\ln D_{\text{ren}} \leftrightarrow 2 \int d^4 x [V_1(\bar{\phi}_b) - V_1(\bar{\phi}_{(\bar{v})})]$ this reproduces (44) coefficient by coefficient.

(c) Numerically, on a real bounce. $N = 2$, $m_i^2 = (2.88792, 7.09173)$, $\mu = \sum_i m_i = 4.362416$; differencing the production counterterm functions in $\ln \mu$:

	finite difference	prediction (44)	rel. dev.
$\partial A_{\text{finite}}^{(1)}/\partial \ln \mu$	+460.8851671171	+460.8851671167	7×10^{-13}
$\partial A_{\text{finite}}^{(2)}/\partial \ln \mu$	-482.3365323	-482.3367053	4×10^{-7}

with moments $\int dr r^3 [U_{\text{int}}'']_{11} = +82.9751$, $\int dr r^3 [U_{\text{int}}'']_{22} = -293.7459$, $\int dr r^3 \text{Tr}[(U_{\text{int}}'')^2] = +3858.694$, and $A_{\text{finite}}^{(1)} = +431.00815$, $A_{\text{finite}}^{(2)} = -297.24707$.

What this pins down

Simultaneously, and end to end against running code: the $1/8 = \Omega_3/16\pi^2$ in $A_{\text{finite}}^{(1)}$; the $-1/(16\pi^2)$ in Π^{finite} ; the $4\pi^2$ in the Hankel kernel; the $-\frac{1}{2}$ and the $\Omega_3/(2\pi)^4$ in $A_{\text{finite}}^{(2)}$; and the statement that the tadpole and the fish are exactly the first two terms of the Coleman–Weinberg expansion in U_{int}'' .

Two readings of (44). (i) The right-hand side is two *local* integrals over the bounce — that is what renormalizability means here: the residual μ -dependence is absorbed by the running of the couplings in the tree action. (ii) The same two numbers are the coefficients of the **logarithmic divergence**. The log has two sources of the same order — the tadpole’s sub-leading ν^{-1} term ($-\frac{1}{4} \int r^3 \text{Tr}[m^2 U_{\text{int}}'']$) and the fish’s leading one ($-\frac{1}{8} \int r^3 \text{Tr}[(U_{\text{int}}'')^2]$). That is no coincidence: the coefficient of a log divergence *is* the coefficient of the μ -running it induces. It is also why $A_{\text{finite}}^{(1)}$ carries a $\ln(m_i^2/\mu^2)$ at all — the tadpole has a log divergence of its own, not just a quadratic one.

Do not conclude that $\ln D_{\text{ren}}$ is μ -independent. Here the two terms nearly cancel (-21.5 per e -fold), but that is an accident of this potential.

13 Checklist

1. Rotate the bounce into the **FV mass eigenbasis**; form $[U''_{\text{int}}(\bar{\phi}_b(r))]_{ij} = [U''(\bar{\phi}_b(r))]_{ij} - m_i^2 \delta_{ij}$ and check it vanishes at large r .
2. For each (n, s^2) : solve the bundled system (6), (8), (9), (10) on both branches. Use log-differences of scaled Bessel functions throughout.
3. Read the Wronskian off at r_{max} : $M = h_n^-(\infty)^T = h_n^+(0)$, $A = (\mathbb{I} + M)^{-1}$. **No transpose in the $\alpha\beta$ slot.**
4. Build $\bar{G}_{(n)} - \bar{G}_{(n),(\bar{v})}$, $\bar{G}_{(n)}^{(1)}$, $\bar{G}_{(n)}^{(2)}$ from (14)–(15) with the measure $r I_{n+1} K_{n+1}$; subtract to get $\delta_{(n)}^{(\geq 3)}$.
5. Integrate over s^2 **per wave** (log-Simpson + low- s^2 rectangle), complete the s^2 tail with (19), then sum over n and close the high- n tail with the Hurwitz zeta.
6. Subtract the dangerous-mode poles at $n = 0, 1$; add $\ln |\omega_{0,0}^2|$; amputate the four zero modes.
7. Compute $A_{\text{finite}}^{(1)}$ from (30) — one radial quadrature.
8. Compute $A_{\text{finite}}^{(2)}$ from (34) — Hankel transform, Feynman parameter, p -integral with an adaptive cutoff. **Mind the $-\frac{1}{2}$.**
9. Assemble (37).
10. **Verify** with (44): difference the two addbacks in $\ln \mu$ and compare against the two local moments. If that fails, a prefactor is wrong.

The procedure in one sentence

Replace the determinant by an s^2 -integral of a traced Green's function; build that Green's function from the h -ODE; peel off its first two Born orders with the linearized h -ODE chain and subtract them pointwise in (n, s^2) so the partial-wave sum converges; then add back, analytically, the finite $\overline{\text{MS}}$ parts of exactly those two orders.